

2 hrs

24 / Mar / 2025

Venue: DJ Exams Hall

EXAMS OFFICE  
USE ONLY

University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS4062A/7074A

Course or topic name(s)  
Paper Number & title

Probabilistic Graphical Models

Examination to be held during the month(s) of

March 2025

Year of study

1

Degrees/Diplomas for which this course is prescribed

BScHons (CS / BDA / CAM), MSc (AI / DS / CS / Robotics/ e-Science)

Faculties presenting candidates

Science

Internal examiner(s)

Prof. Ritesh Ajoodha  
x-76188

External examiner(s)

Prof. JH Van Vuuren (Stellenbosch Uni.)

Special materials

Formula sheet and non-programmable calculator permitted

Time allowance

2 hours

Instructions to candidates

See next page



## Student Details

Please fill in the details below with a pen:

**Student Number:** \_\_\_\_\_

**Student Name:** \_\_\_\_\_

**Exam Instructions:** Please answer all questions in this closed book test. A total of 60 marks are available, which corresponds to 100%. This exam has 21 pages.

### Mark Table:

| Component                                   | Marks    | Grade |
|---------------------------------------------|----------|-------|
| Question 1: Problem Solving                 | 10 Marks |       |
| Question 2: MCQs                            | 10 Marks |       |
| Question 3: Bayesian and Markov Networks    | 10 Marks |       |
| Question 4: Local Probability Models        | 10 Marks |       |
| Question 5: Template-based Models           | 10 Marks |       |
| Question 6: Exact and Approximate Inference | 10 Marks |       |

**Question 1                      Problem Solving using PGMs                      [10 Marks]**

1.1 Patients are dying because chronic diseases like diabetes, cardiovascular disorders, and respiratory illnesses are often diagnosed too late, limiting treatment effectiveness. Early detection could save lives, but healthcare providers struggle to identify high-risk individuals due to incomplete records, complex risk factor interactions, and genetic and environmental influences. You have access to a 20-year longitudinal patient dataset with the following key features:

- (a) Patient Demographics: *Age, gender, ethnicity, socioeconomic status, and lifestyle factors (e.g., smoking, diet, physical activity).*
- (b) Medical History: *Previous diagnoses, family history of chronic illnesses, medication usage, and past hospitalizations.*
- (c) Physiological Data: *Vital signs (e.g., blood pressure, heart rate, BMI), lab test results, and imaging reports.*
- (d) External Factors: *Environmental conditions, access to healthcare facilities, and regional disease outbreaks.*

Design a Probabilistic Graphical Model (PGM) framework that can predict a patient's likelihood of developing chronic diseases such as diabetes, cardiovascular disorders, or respiratory illnesses based on their historical and real-time health data.

Your model should account for uncertainty in medical data, correlations among risk factors, and missing values while providing actionable insights for healthcare practitioners.

You will be marked on the following criteria:

- (a) Problem representation using PGMs [4]
- (b) Inference and Learning approach used [3]
- (c) Limitations, challenges, and considerations [3]

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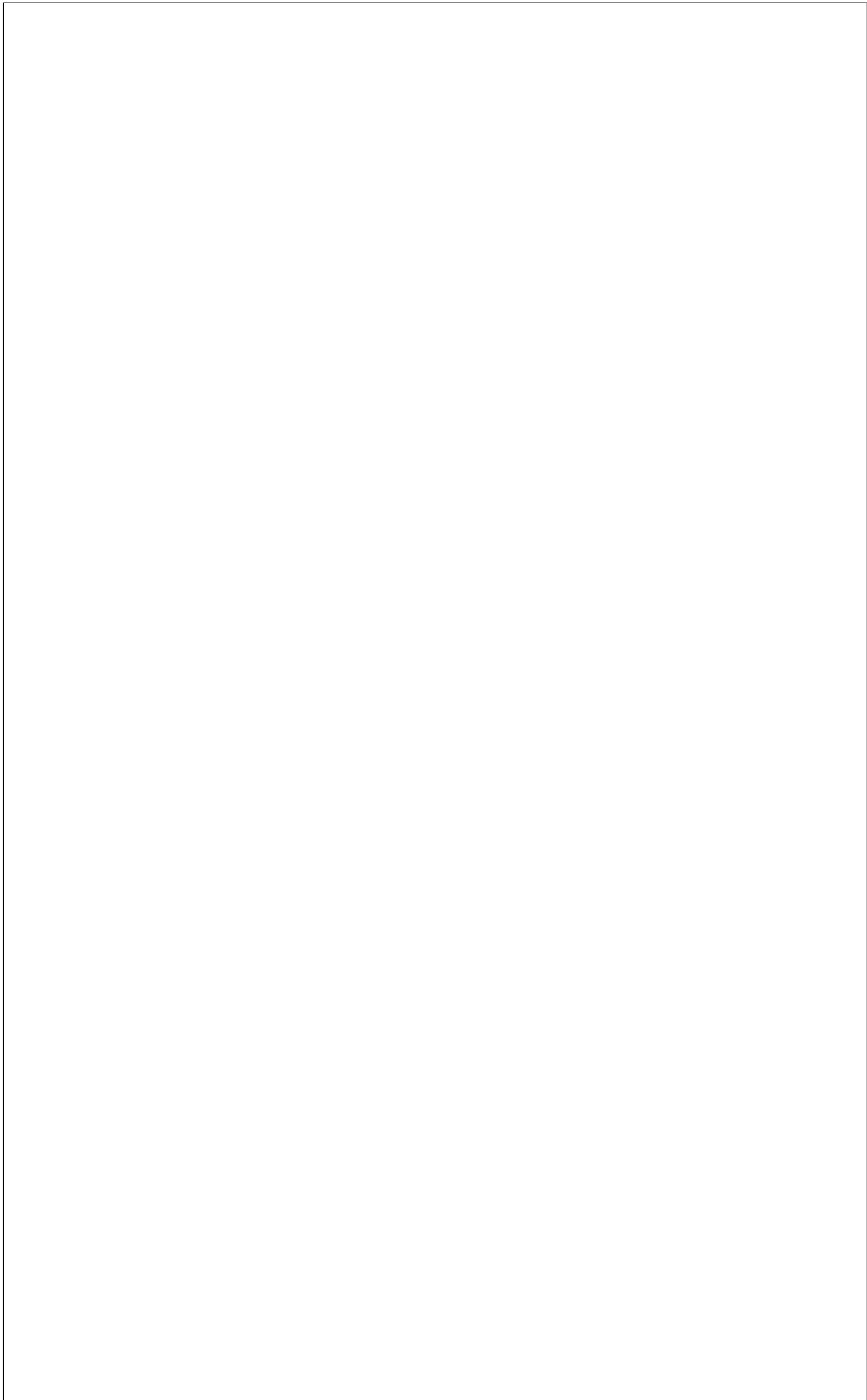
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**Question 2                      Multiple Choice Questions                      [10 Marks]**

2.1. For each of the following MCQ questions, circle the correct answer label.

2.1.1. What is the primary advantage of using Probabilistic Graphical Models (PGMs) in real-world applications? [2]

- (a) PGMs assume all variables are independent, simplifying computation.
- (b) PGMs provide a structured way to model uncertainty and dependencies in complex systems. (ANSWER)
- (c) PGMs require complete data and cannot handle missing values.
- (d) PGMs always assume a linear relationship between variables.

2.1.2. Which of the following correctly describes the relationship between Bayesian Networks and Markov Networks? [2]

- (a) Bayesian Networks and Markov Networks both represent undirected dependency structures.
- (b) Bayesian Networks use directed edges to model causal relationships, while Markov Networks use undirected edges to capture correlations. (ANSWER)
- (c) Markov Networks can always be converted into Bayesian Networks without loss of information.
- (d) Bayesian Networks cannot represent conditional independence constraints.

2.1.3. Which statement is true regarding independence properties in PGMs? [2]

- (a) Bayesian Networks always capture the full set of conditional independencies present in the data.
- (b) Bayesian Networks can capture  $P(X \perp Y \mid Z)$  but may introduce extra dependencies due to additional edges.
- (c) Markov Networks encode  $P(X \perp Y \mid Z)$  using factorized clique potentials without requiring directed edges. (ANSWER)
- (d) Bayesian Networks and Markov Networks always encode the same independence properties.

2.1.4. What is a key challenge when performing inference in PGMs? [2]

- (a) Exact and Approximate inference in general PGMs is  $\mathcal{NP}$ -hard. (ANSWER)
- (b) Inference in PGMs always has polynomial time complexity, making it computationally efficient.
- (c) Probabilistic inference in PGMs requires converting them into deterministic models for tractability.
- (d) Message passing algorithms such as belief propagation always provide exact solutions, regardless of network complexity.

2.1.5. In Temporal PGMs, what does the Markov Assumption imply? [2]

- (a) The future state depends on all past states in the system.

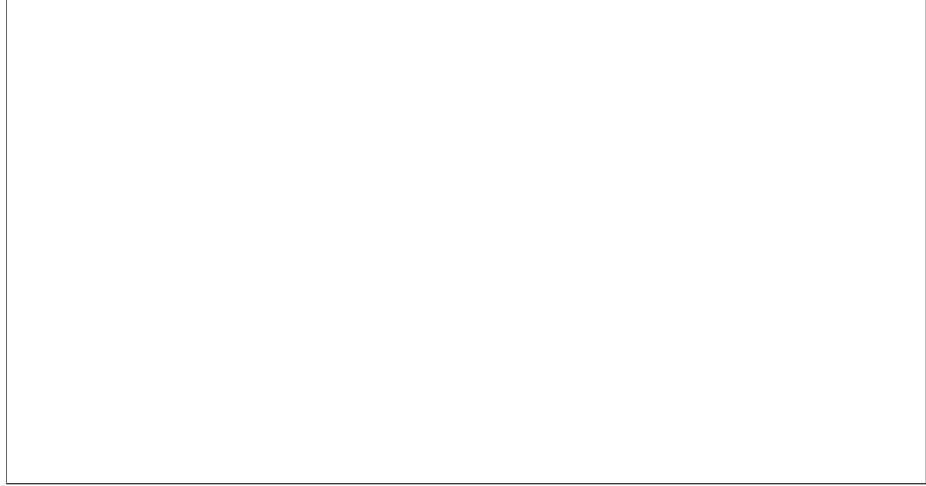
- (b) The future state depends only on the present state, not the entire past. (ANSWER)
- (c) The Markov assumption enforces complete independence between time steps.
- (d) The assumption ensures that every state transition is deterministic.





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- ii. List all of the conditional independence assumptions associated with the Bayesian network structure in [Figure 1](#). [2]
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- iii. Given that  $\mathcal{G}$  is now an undirected Markov network, preserving the deduced properties. Use the structure learning algorithm in class to draw the Markov network corresponding to the minimal I-map based on the observed independence assumptions, using the variable ordering  $\prec = \{D, C, B, A\}$ . [2]
- 



3.3 Use the below factor graph,  $\mathcal{H}$ , to answer the following questions.

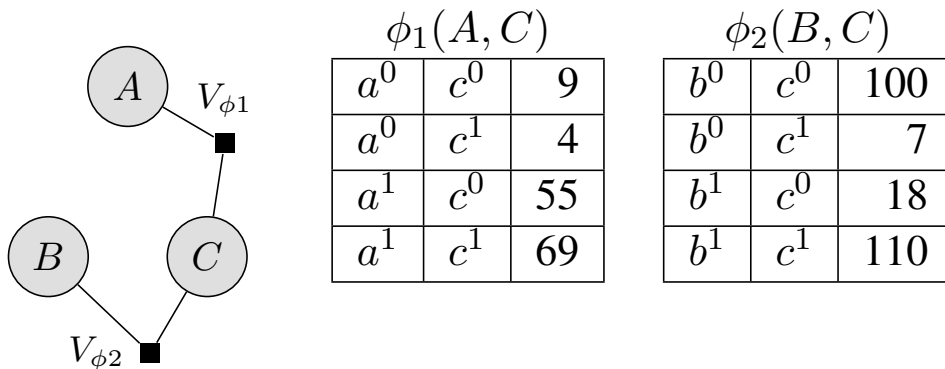


Figure 2: A factor graph with associated factors.

- i. Write the Gibbs distribution of  $\mathcal{H}$  using maximal clique potentials,  $\Phi$ . [1]

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- ii. Compute the factor product  $\psi_3(A, B, C) = \phi_1(A, C) \times \phi_2(B, C)$ . [1]

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## Question 4      Local Probability Models      [10 Marks]

- 4.1. Consider the Tabular-CPD,  $\mathcal{T}$ , shown in Figure 3, and use it to answer the following questions.

| $A$   | $B$   | $C$   | $P(A, B, C)$ |
|-------|-------|-------|--------------|
| $a^0$ | $b^0$ | $c^0$ | 0.13         |
| $a^0$ | $b^0$ | $c^1$ | 0.15         |
| $a^0$ | $b^1$ | $c^0$ | 0.1          |
| $a^0$ | $b^1$ | $c^1$ | 0.14         |
| $a^1$ | $b^0$ | $c^0$ | 0.2          |
| $a^1$ | $b^0$ | $c^1$ | 0.04         |
| $a^1$ | $b^1$ | $c^0$ | 0.2          |
| $a^1$ | $b^1$ | $c^1$ | 0.04         |

Figure 3: A Tabular-CPD denoted  $\mathcal{T}$

- i. Name one advantage and one disadvantage of using a Tree CPDs over a Rule CPDs? [2]

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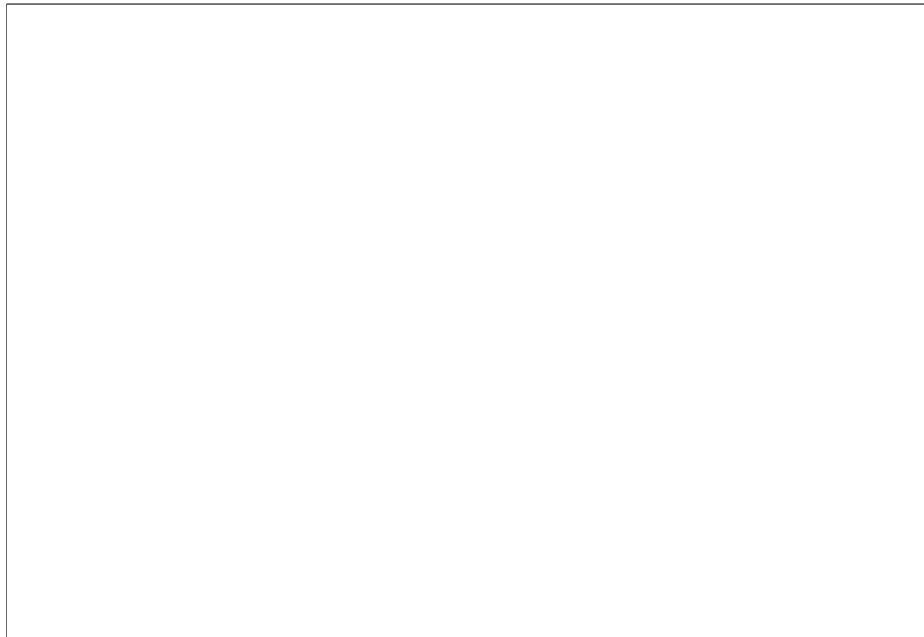


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- ii. Draw  $\mathcal{T}$  as a (normalised) Tree-CPD which specifies the joint distribution between the three variables A, B, and C. [4]



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- iii. Name one context specific independence assumption,  $(\mathbf{X} \perp_c \mathbf{Y} \mid \mathbf{Z})$ , that holds in  $\mathcal{T}$ ? [1]

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- iv. List all of the rules,  $\rho = \langle \mathbf{c}; p \rangle$ , which together give you the Rule-CPD that hold in  $\mathcal{T}$ . [3]

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## Question 5      Template-based Models      [10 Marks]

5.1. Use the below parametrization of a hidden Markov model (HMM),  $\lambda$ , to answer the following questions.

- i. Number of hidden states: 3.
- ii. There are 4 observable symbols:  $\{0,1,2,3\}$ .
- iii. Initial state probabilities:  $\pi_1 = 0.7, \pi_2 = 0.1, \pi_3 = 0.2$
- iv. The transition probability matrix is:  $\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 0.5 & 0.3 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}$
- v. The observation model is provided by the following matrix:  $\begin{pmatrix} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.3 & 0.2 & 0.2 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{pmatrix}$
- i. Draw the structure of  $\mathcal{H}$  with the associated probabilities labelled at the correct edges. Remember to label each node. [2]

- ii. Calculate the initial state probabilities for the forward variables  $\alpha_1(1)$ ,  $\alpha_1(2)$ , and  $\alpha_1(3)$  using  $\lambda$  for the sequence  $O = \{2, 3\}$ . [2]

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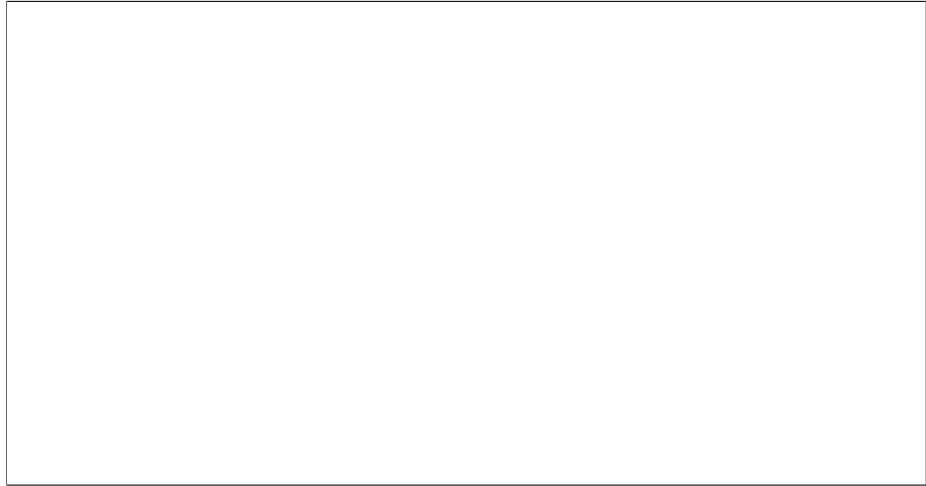
- iii. Compute the forward variables  $\alpha_2(1)$ ,  $\alpha_2(2)$ , and  $\alpha_2(3)$  using  $\lambda$  for the first induction step using the initial state probabilities from the previous question. [4]

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- iv. Using the previous question, calculate the  $P(O \mid \lambda)$  for the sequence  $O = \{2, 3\}$ . [2]
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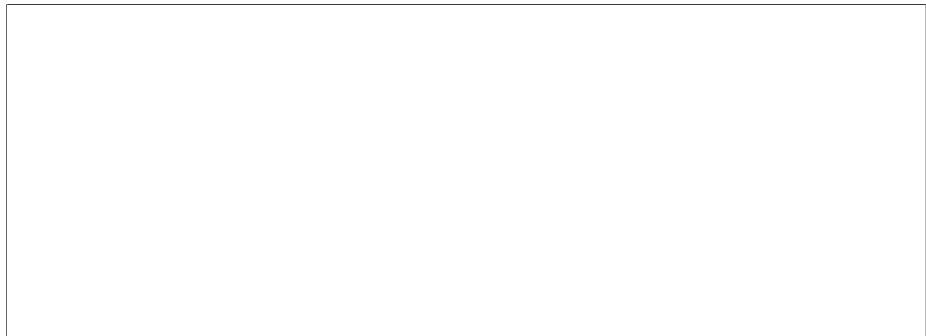


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- ii. Draw the induced graph  $\mathcal{I}_{\mathcal{B}, \prec}$  which results in this ordering of variable elimination. [1]



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- iii. Draw the clique tree for  $\mathcal{I}_{\mathcal{B}, \prec}$  [1]



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END OF EXAM

[illegible]

[illegible]

# Probabilistic Graphical Models

## Formula Sheet

### Probability Theory

**Chain Rule for Probabilities:**

$$P(X_1, \dots, X_n) = P(X_1) \dots P(X_n | X_1, X_{n-1})$$

**Bayes Rule:**

$$P(\alpha | \beta) = \frac{P(\beta | \alpha)}{P(\alpha)P(\beta)}$$

**Probability Density Function:**  $p : \mathbb{R} \rightarrow \mathbb{R}$  is a

probability density function (PDF) for  $\mathcal{X}$  if it is a non-negative integrable function such that:

$$\int_{\mathcal{V}al(\mathcal{X})} p(x) dx = 1.$$

**Uniform Distribution:**  $X \sim \text{Unif}[a, b]$  if it has the PDF:

$$p(x) = \begin{cases} \frac{1}{b-a} & b \geq x \geq a \\ 0 & \text{otherwise.} \end{cases}$$

**Gaussian Distribution:**  $X$  has a Gaussian

distribution:  $X \sim \mathcal{N}(\mu; \sigma^2)$  if it has the PDF:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

**Joint Density Function:** Let  $P$  be a joint

distribution over  $X_1, \dots, X_n$ . A function

$p(x_1, \dots, x_n)$  is a joint density function of  $X_1, \dots, X_n$  if:

1.  $p(x_1, \dots, x_n) \geq 0 \forall x_1, \dots, x_n \in X_1, \dots, X_n$ .
2.  $p$  is integratable.
3. For any choice of  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$ :

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) \\ = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} p(x_1, \dots, x_n) dx_1 \dots dx_n$$

**Conditional Density Function:** Suppose you would like to condition over the event:

$$x - \epsilon \leq X \leq x + \epsilon.$$

Then

$$P(Y | x) = \lim_{\epsilon \rightarrow 0} P(Y | x - \epsilon \leq X \leq x + \epsilon).$$

If there is a continuous joint density function  $p(x, y)$  then

$$P(a \leq Y \leq b | x - \epsilon \leq X \leq x + \epsilon) \\ = \frac{P(a \leq Y \leq b, x - \epsilon \leq X \leq x + \epsilon)}{P(x - \epsilon \leq X \leq x + \epsilon)} = \frac{\int_a^b \int_{x-\epsilon}^{x+\epsilon} p(x', y) dy dx'}{\int_{x-\epsilon}^{x+\epsilon} p(x') dx'}$$

**Expectation** of  $X$  under  $P$  is:

$$\mathbb{E}_P[X] = \sum_{\mathcal{X}} x.P(x).$$

**Expectation if  $\mathbf{X}$  is Continuous:**

$$\mathbb{E}_P[X] = \int x.p(x) dx.$$

**Linearity of Expectation:**

$$\mathbb{E}_P[X + Y] = \mathbb{E}_P[X] + \mathbb{E}_P[Y].$$

**Conditional Expectation:**

$$\mathbb{E}_P[X | \mathbf{y}] = \sum_x x.P(x | \mathbf{y}).$$

**Variance of  $\mathbf{X}$ :**

$$\mathbb{V}ar_P[X] = \mathbb{E}_P[(X - \mathbb{E}_P[X])^2].$$

**Standard Deviation:**

$$\sigma_X = \sqrt{\mathbb{V}ar_P[X]}.$$

**Expectation and Variance of Gaussian**

**distribution**  $X \sim \mathcal{N}(\mu; \sigma^2)$ , then  $\mathbb{E}[X] = \mu$  and

$$\mathbb{V}ar[X] = \sigma^2.$$

### Graph Theory

A **Graph** is a data structure  $\mathcal{K} = (\mathcal{X}, \mathcal{E})$  consisting

of a set of nodes, denoted  $\mathcal{X} = X_1, \dots, X_n$ , and

edges, denoted  $\mathcal{E}$ .

**Induced Subgraph:** Let  $\mathcal{K} = (\mathcal{X}, \mathcal{E})$ , and  $\mathbf{X} \in \mathcal{X}$ ,

then an induced subgraph, denoted  $\mathcal{K}[\mathbf{X}]$  is a graph

$(\mathbf{X}, \mathcal{E}')$  where  $\mathcal{E}'$  are all the edges  $X \rightleftharpoons Y \in \mathcal{E}'$  such

that  $X, Y \in \mathbf{X}$ .

**Complete Graph (Clique):** A subgraph over  $\mathbf{X}$  is

complete if every two nodes in  $\mathbf{X}$  are connected by

some edge. The set  $\mathbf{X}$  is called a clique. A clique  $\mathbf{X}$

is maximal if for any superset of nodes  $\mathbf{Y} \supset \mathbf{X}$ ,  $\mathbf{Y}$  is

not a clique.

**Upward Closure:** A subset of nodes  $\mathbf{X} \in \mathcal{X}$  is

upwardly closed in  $\mathcal{K}$  if, for any  $\mathbf{X} \in \mathcal{X}$ , we have that

the Boundary  $\mathbf{x} \subset \mathbf{X}$ . We define upward closure of  $\mathbf{X}$

to be the minimally upward closed subset  $\mathbf{Y}$  that

contains  $\mathbf{X}$ .

**Topological ordering:** An ordering of the nodes

$X_1, \dots, X_n$  is a topological ordering if when we have

$(X_i \rightarrow X_j) \in \mathcal{E}$ , then  $i < j$ .

**Chordal Graph:** Let  $X_1 - X_2 - \dots - X_k - X_1$  be a

loop in a graph. A chord in a loop is an edge

connecting  $X_i$  and  $X_j$  for two nonconsecutive nodes

$X_i, X_j$ . An undirected graph  $\mathcal{H}$  is said to be chordal

if and loop  $X_1 - X_2 - \dots - X_k - X_1$  for  $k > 4$  has a

chord. A directed graph  $\mathcal{K}$  is said to be chordal if its

underlying undirected graph is chordal.

### Bayesian Networks

**Naïve Bayes:**

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i | C)$$

**Bayesian Network:**

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Pa}_{X_i}^G)$$

**Deterministic CPD:**  $f : \mathcal{V}al(P_{a_X}) \mapsto \mathcal{V}al(X)$  s.t.:

$$P(x | pa_x) = \begin{cases} 1 & \text{if } x = f(pa_x) \\ 0 & \text{if } x \text{ otherwise} \end{cases}$$

**Time Granularity Assumption:**

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} | \mathcal{X}^{(0:t)})$$

**Markov Assumption:**

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} | \mathcal{X}^{(t)})$$

**Time Invariance Assumption:**

$$P(\mathcal{X}^{(t+1)} = \xi' | \mathcal{X}^{(t)} = \xi) = P(\mathcal{X}' = \xi' | \mathcal{X} = \xi)$$

**Two-TBN:**

$$P(\mathcal{X}' | \mathcal{X}) = P(\mathcal{X}' | \mathcal{X}_t) = \prod_{i=1}^n P(X'_i | P_{a_{X'_i}})$$

**Linear Dynamical Systems:**

$$P(\mathbf{X}^{(t)} | \mathbf{X}^{(t-1)}) = \mathcal{N}(\mathbf{A}\mathbf{X}^{(t-1)}; Q)$$

$$P(O^{(t)} | \mathbf{X}^{(t)}) = \mathcal{N}(H\mathbf{X}^{(t)}; R)$$

**Gibbs Distribution:** A distribution  $P_\Phi$  is a Gibbs

distribution parameterised by a set of factors

$\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$  if it is defined as:

$$P_\Phi(X_1, \dots, X_n) = \frac{1}{Z} P_\Phi(X_1, \dots, X_n)$$

### Inference

**Inference:**

$$P(\mathbf{Y} | \mathbf{E} = \mathbf{e}) = \frac{P(\mathbf{Y}, \mathbf{e})}{P(\mathbf{e})} = \frac{\sum_{\mathbf{w}} P(\mathbf{y}, \mathbf{e}, \mathbf{w})}{\sum_{\mathbf{y}, \mathbf{w}} P(\mathbf{e})}$$

**Sum-Product Message Passing:**

$$\delta_{i \rightarrow j} = \sum_{\mathbf{C}_i - \mathbf{s}_{i,j}} (\psi_i \times \prod_{k \in (N b_i - \{j\})} \delta_{k \rightarrow i})$$

**Tree Calibration:**

$$\sum_{\mathbf{C}_i - \mathbf{s}_{i,j}} \beta_i(\mathbf{C}_i) = \sum_{\mathbf{C}_j - \mathbf{s}_{i,j}} \beta_j(\mathbf{C}_j)$$

**Graph Calibration:**

$$\sum_{\mathbf{C}_i - \mathbf{s}_{i,j}} \beta_i = \sum_{\mathbf{C}_j - \mathbf{s}_{i,j}} \beta_j$$

**MAP:**

$$\text{MAP}(\mathbf{Y} = \mathbf{y} | \mathbf{E} = \mathbf{e})$$

$$= \text{argmax}_{\mathbf{y}} P(\mathbf{Y} = \mathbf{y} | \mathbf{E} = \mathbf{e})$$

**Convergence Bound:**

$$\mathbb{E}_{\mathcal{D}}(f) = \frac{1}{M} \sum_{m=1}^M f(\xi[m]).$$

**Hoeffding Bound:**

$$P_{\mathcal{D}}(\hat{P}(\mathbf{y}) \notin [P(\mathbf{y}) - \epsilon, P(\mathbf{y}) + \epsilon]) \leq 2e^{-2M\epsilon^2}$$

**Chernoff Bound:**

$$P_{\mathcal{D}}(\hat{P}(\mathbf{y}) \notin [P(\mathbf{y})(\pm\epsilon)]) \leq 2e^{-MP(\mathbf{y})\epsilon^2/3}$$

$$M \geq 3 \frac{\ln(2/\delta)}{P(\mathbf{y})\epsilon^2}.$$

### Likelihood Weighting:

$$\hat{P}_{\mathcal{D}}(\mathbf{y} \mid \mathbf{e}) = \frac{\sum_{m=1}^M w[m] \mathbb{1}\{\mathbf{y}[m] = \mathbf{y}\}}{\sum_{m=1}^M w[m]}.$$

### MCMC Sampling:

$$P^{(t+1)}(\mathbf{X}^{(t+1)} = \mathbf{x}') =$$

$$\sum_{\mathbf{x} \in \text{Val}(\mathbf{X})} P^{(t)}(\mathbf{X}^{(t)} = \mathbf{x}) \mathcal{T}(\mathbf{x} \rightarrow \mathbf{x}')$$

### Stationary Distribution:

$$\pi(\mathbf{X} = \mathbf{x}') = \sum_{\mathbf{x} \in \text{Val}(\mathbf{X})} \pi(\mathbf{X} = \mathbf{x}) \mathcal{T}(\mathbf{x} \rightarrow \mathbf{x}')$$

### Detailed Balance Equation:

$$\pi(x) \mathcal{T}(x \rightarrow x') = \pi(x') \mathcal{T}(x' \rightarrow x)$$

### Acceptance Probability:

$$\mathcal{A}(x \rightarrow x') = \min\left[1, \frac{\pi(x') \mathcal{T}^Q(x' \rightarrow x)}{\pi(x) \mathcal{T}^Q(x \rightarrow x')}\right].$$

### Metropolis-Hastings Acceptance Probability:

$$\mathcal{A}(x_{-i}, x_i \rightarrow x_{-i}, x'_i) = \min\left[1, \frac{P_{\theta}(x'_i, x_{-i}) \mathcal{T}_{\theta}^Q(x_{-i}, x'_i \rightarrow x_{-i}, x_i)}{P_{\theta}(x_i, x_{-i}) \mathcal{T}_{\theta}^Q(x_{-i}, x_i \rightarrow x_{-i}, x'_i)}\right].$$

## Learning

### Relative Entropy:

$$\mathbb{D}(P^* \parallel \tilde{P}) = \mathbb{E}_{\xi \sim P^*} \left[ \log \left( \frac{P^*(\xi)}{\tilde{P}(\xi)} \right) \right],$$

### Negative Empirical Log-loss:

$$\log P(\mathcal{D} : \mathcal{M}) = \sum_{m=1}^M \log P(\xi[m] : \mathcal{M}).$$

### Bayesian Parameter Estimation:

$$P(\theta \mid x[1], \dots, x[M]) = \frac{P(x[1], \dots, x[M] \mid \theta) P(\theta)}{P(x[1], \dots, x[M])}$$

### Expected Sufficient Statistics:

$$\bar{M}_{\theta}[\mathbf{y}] = \sum_{m=1}^M \sum_{\mathbf{h}[m] \in \text{Val}(\mathbf{H}[m])} Q(\mathbf{h}[m]) \mathbb{1}\{\xi[m]\langle \mathbf{Y} \rangle = \mathbf{y}\}$$

### Maximisation of Expected Parameter:

$$\tilde{\theta}_{d^1|c^0} = \frac{M_{\theta}(d^1, c^0)}{M_{\theta}(c^0)}$$

### Bayesian Clustering:

$$\bar{M}_{\theta}[c] = \frac{\bar{M}_{\theta}[c]}{M}$$

$$\bar{M}_{\theta}[x_i \mid c] = \frac{\bar{M}_{\theta}[x_i, c]}{\bar{M}_{\theta}[c]}$$

### K-means Clustering:

$$c[m] = \text{argmax}_c P(c \mid x[m], \theta^t)$$

### Hypothesis Testing:

$$d_{\mathbb{H}}(\mathcal{D}) = \sum_{x,y} \frac{M[x,y]}{x,y} \log \frac{M[x,y]/M}{M[x]/M \cdot M[y]/M}$$

$$R_{d,t}(\mathcal{D}) \begin{cases} \text{Accept} & \text{if } d(\mathcal{D}) \leq t \\ \text{Reject} & \text{if } d(\mathcal{D}) > t \end{cases}$$

$$\text{p-value}(t) = P(\{\mathcal{D} : d(\mathcal{D}) > t\} \mid H_0, M)$$

### Likelihood:

$$\mathbb{I}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i; Pa_{\mathbf{X}_i}^{\mathcal{G}})$$

$$= \sum_{\mathbf{u}_i} \sum_{\mathbf{x}_i} \hat{P}(x_i, \mathbf{u}_i) \log \frac{\hat{P}(x_i, \mathbf{u}_i)}{\hat{P}(x_i) \hat{P}(\mathbf{u}_i)}$$

### Entropy:

$$\mathbb{H}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i) = \sum_{x_i} \hat{P}(x_i) \log \frac{1}{\hat{P}(x_i)}$$

### Bayesian Structure Learning:

$$P(\mathcal{G} \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \mathcal{G}) P(\mathcal{G})}{P(\mathcal{D})}$$

$$\text{score}_{\mathcal{B}}(\mathcal{G} : \mathcal{D}) = \log P(\mathcal{D} \mid \mathcal{G}) + \log P(\mathcal{G})$$

$$P(\mathcal{D} \mid \mathcal{G}) = \int_{\Theta_{\mathcal{G}}} P(\mathcal{D} \mid \theta_{\mathcal{G}}, \mathcal{G}) P(\theta_{\mathcal{G}} \mid \mathcal{G}) d\theta_{\mathcal{G}}$$

### Marginal Likelihood for Binomials:

$$P(x[1], \dots, x[M])$$

$$= P(x[1]) \cdot \dots \cdot P(x[M] \mid x[1], \dots, x[M-1])$$

### Marginal Likelihood for Multinomials:

$$P(x[1], \dots, x[M]) = \frac{\Gamma(\alpha)}{\Gamma(\alpha+M)} \cdot \prod_{i=1}^k \frac{\Gamma(\alpha_i + M[x^i])}{\Gamma(\alpha_i)}$$

### Bayesian Score:

$$P(\mathcal{D} \mid \mathcal{G}) = \prod_i \prod_{\mathbf{u}_i \in \text{Val}(Pa_{\mathbf{X}_i}^{\mathcal{G}})} \frac{\Gamma(\alpha_{\mathbf{X}_i \mid \mathbf{u}_i}^{\mathcal{G}})}{\Gamma(\alpha_{\mathbf{X}_i \mid \mathbf{u}_i}^{\mathcal{G}} + M[\mathbf{u}_i])}.$$

$$\prod_{\mathbf{x}_i^j \in \text{Val}(\mathbf{X}_i)} \left[ \frac{\Gamma(\alpha_{\mathbf{X}_i \mid \mathbf{u}_i}^{\mathcal{G}} + M[x_i^j, \mathbf{u}_i])}{\Gamma(\alpha_{\mathbf{X}_i \mid \mathbf{u}_i}^{\mathcal{G}})} \right]$$

### BIC Score:

$$\text{score}_{\text{BIC}}(\mathcal{G} : \mathcal{D}) =$$

$$M \sum_{i=1}^n \mathbb{I}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i; Pa_{\mathbf{X}_i}^{\mathcal{G}}) - \frac{\log M}{2} \dim[\mathcal{G}]$$

### Decomposability:

$$\text{score}(\mathcal{G} : \mathcal{D}) = \sum_i \text{FamScore}(\mathbf{X}_i \mid Pa_{\mathbf{X}_i}^{\mathcal{G}} : \mathcal{D})$$

### Tree weight:

$$w_{i \rightarrow j} = \text{FamScore}(\mathbf{X}_i \mid \mathbf{X}_j : \mathcal{D}) -$$

$$\text{FamScore}(\mathbf{X}_i : \mathcal{D})$$

### Learning Graphs:

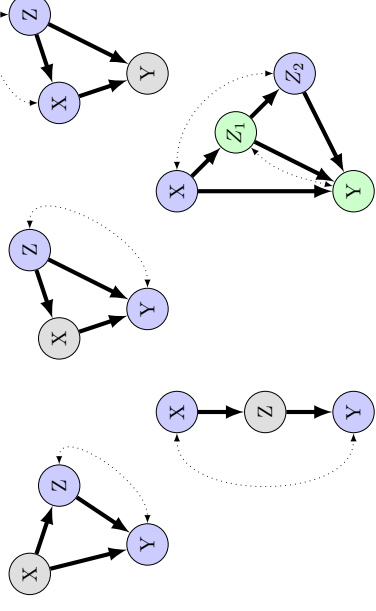
$$\mathcal{G}^* = \text{argmax}_{\mathcal{G} \in \mathcal{G}} \text{score}(\mathcal{G} : \mathcal{D})$$

## Causality

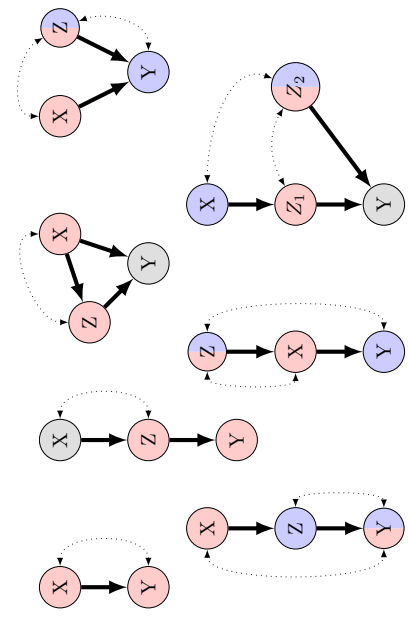
### Intervention Query:

$$P_{\mathcal{C}}(\mathbf{Y} \mid do(z), \mathbf{x}) = P_{\mathcal{C}_{z=x}}(\mathbf{Y} \mid \mathbf{x})$$

### Identifiable when $P(Y \mid do(X))$ :



### Not Identifiable when $P(Y \mid do(X))$ :



### Learning with Intervention Data:

$$P(\xi \mid do(\mathbf{Z} := \mathbf{z}), \mathcal{C}) = \prod_{X_i \notin \mathbf{Z}} P(x_i \mid \mathbf{u}_i)$$

### Sufficient Statistics (Intervention Data):

$$M[x_i; \mathbf{u}_i] =$$

$$\sum_{m: X_i \notin \mathbf{Z}[m]} \mathbb{1}\{X_i[m] = x_i, Pa_{X_i}[m] = \mathbf{u}_i\}$$

### Likelihood of Data (Intervention):

$$L(\mathcal{C} : \mathcal{D}) = \prod_{i=1}^n \prod_{x_i \in \text{Val}(X_i), \mathbf{u}_i \in \text{Val}(Pa_{X_i})} \theta_{x_i | \mathbf{u}_i}^{M[x_i; \mathbf{u}_i]}$$

## Decision Theory

### Expected Utility:

$$\text{EU}[D[a]] = \sum_{\mathbf{x}} P(\mathbf{x} \mid a) U(\mathbf{x}, a)$$

### Maximum Expected Utility:

$$a^* = \text{argmax}_a \text{EU}[D[a]]$$

$$= \text{argmax}_a \sum_{\mathbf{x}} P(\mathbf{x} \mid a) U(\mathbf{x}, a)$$

### Expected Utility with Information:

$$\text{EU}[D[\delta_A]] = \sum_{\mathbf{x}, a} P_{\delta_A}(\mathbf{x}, a) U(\mathbf{x}, a)$$

### Maximal Expected Utility (MEU) Strategy:

$$\text{argmax}_{\delta_{D_1}, \dots, \delta_{D_k}} \text{EU}[\mathcal{Z}[\delta_{D_1}, \dots, \delta_{D_k}]]$$

### Value of Information:

$$\text{VPI}(A \mid X) := \text{MEU}(D_{X \rightarrow A}) - \text{MEU}(D)$$